BSH/S-24/CS/DS/AI2T005/02

JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

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Education to Eternity

(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU& RTMNU

B.Tech / ~~M.Tech~~ / ~~MBA~~ / ~~MCA~~ Summer / ~~Winter~~ - 2024

(ESE / ~~Make-up~~ / ~~Supplementary~~) Examination

Branch: CS/DS/AI **Sem:-II**

Subject: Energy and Environment Engineering

Subject Code: CS/DS/AI2T005 **Marks: 100**

Date of Exam: 07/06/2024 **Time: 3 Hrs.**

Instructions to the Students:

1. **Read the question paper carefully (Branch, Semester, Scheme) before attempting the questions.**
2. **All the questions carry marks as indicate.**
3. **Use of Programmable Calculator is prohibited.**
4. **Assume suitable data whenever necessary.**
5. **Illustrate your answer whenever necessary with the help of neat sketches.**
6. **Don't use red pen for writing the answers.**

CO :- At the end of the course students will be able to

CO1. Identify major factors that would affect the atmosphere and the climate change and energy generation system

CO2. Understand the importance of energy efficiency and formulate energy conservation strategy for sustainable environment

CO3. Apply advanced level knowledge, techniques, skills and modern tools in the field of Energy and Environmental Engineering

CO4. Distinguish the different energy generation systems and their environmental impacts.

CO5. Respond to global policy initiatives and meet the emerging challenges with sustainable technological solutions in the field of energy and environment.

CO6. Manage projects related to Energy and Environmental Engineering in multidisciplinary environments

SECTION A

Q. No.	Solve the two words or two sentences (All Questions are compulsory).	Bloom's Level and CO	Marks
Q.1	Define noise Pollution.	1/CO1	2
Q.2	Enlist various water pollutants with example.	1/CO1	2
Q.3	What is acid rain?	2/CO2	2
Q.4	List chemical characteristics of water.	1/CO1	2
Q.5	What is biomedical waste?	1/CO1	2
Q.6	Name any four components of Rooftop method of rainwater harvesting.	2/CO2	2
Q.7	Define Biomass	1/CO1	2
Q.8	List examples of primary and secondary air pollutant.	2/CO2	2

Q.9	What do you understand by incineration?	1/CO1	2
Q.10	What are conventional energy sources?	2/CO2	2
Q.11	Define energy efficiency.	1/CO1	2
Q.12	Relate the role of air preheater in a steam power plant	1/CO1	

SECTION B

Q.No.	Solve the following Descriptive Type Questions (All Questions are compulsory).	Bloom's Level and CO	Marks
Q.13	Explain renewable energy sources. Also mention its advantages and disadvantages.	3/CO3	4
Q.14	Describe different advantages and disadvantages of nuclear power plant.	4/CO4	4
Q.15	Illustrate the sources of air pollution.	4/CO4	4
Q.16	Describe site selection criteria for Hydro power plant	3/CO3	4
Q.17	Explain noise pollution and its control measures.	4/CO4	4
Q.18	Outline different health problems caused by water pollution	3/CO3	4

SECTION C

Q. No.	Solve the following Descriptive Type Questions (Solve any 04 out of 06).	Bloom's Level and CO	Marks
Q.19	Describe rain water harvesting and its types.	5/CO5	7
Q.20	Innumerate the various components used in nuclear power plant with neat diagram.	4/CO4	7
Q.21	Explain methods of disposal of solid waste in detail.	6/CO6	7
Q.22	Utilize your knowledge to explain methods of water conservation.	4/CO4	7
Q.23	Explain working of Wind turbine with the help of neat sketch.	6/CO6	7
Q.24	Justify control measures to be taken to reduce air pollution.	5/CO5	7

SECTION D

Q. No.	Solve the following Descriptive Type Questions (Solve any 02 out of 03).	Bloom's Level and CO	Marks
Q.25	Explain with the help of neat sketch, working of steam power plant. Also write its advantages and disadvantages	6/CO6	12
Q.26	Explain biogas .Also explain working of biogas power plant with neat sketch and advantages as well as limitations.	6/CO6	12
Q.27	Analyze and explain the components of tidal power plant with a neat sketch. Write advantages as well as limitations of tidal power plant.	4/CO4	12